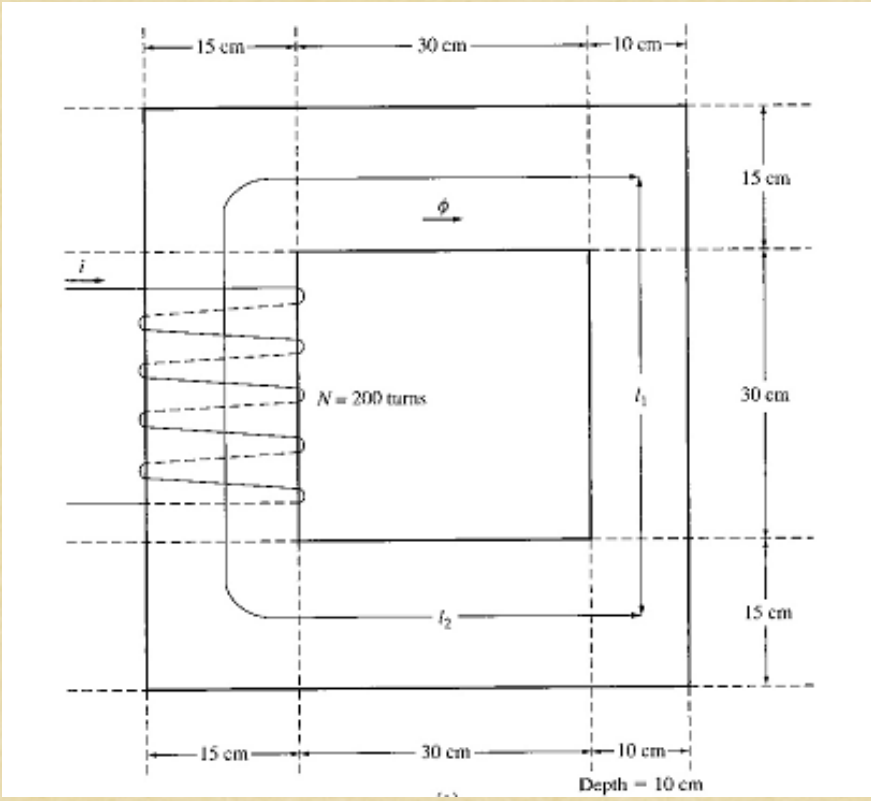
EE2005: Electrical Machines
Tutorial 1

1. A ferromagnetic core is shown. Three sides of this core are of uniform width, while the fourth side is somewhat thinner. The depth of the core (into the page) is 10 cm, and the other dimensions are shown in the figure. There is a 200 turn coil wrapped around the left side of the core. Assuming relative permeability of 2500, how much flux will be produced by a 1A input current?



2. A toroidal ring has a circular cross section of 3 cm in diameter and a mean circumference of 80 cm. The ring is uniformly wound with a coil of 600 turns.

   (a) Estimate the current in the coil required to produce a flux of 0.5 mWb in the ring, when

      (i) The entire core region is replaced by air

      (ii) A core made of solid cast steel of relative permeability 3000 is used.

   (b) If a saw cut creates a 2 mm air gap in the ring, find approximately the air gap flux produced by the current obtained in (a - ii). (Neglect magnetic leakage and fringing)

   (c) Find the current required in the above case to produce the same flux in the air gap as in (a - ii).

(b) $\mathcal{R}_g = \frac{2\times10^{-3}}{4\pi\times10^{-7}\times\pi(3\frac{1}{2}\times10^{-2})^2} = 2.25\times10^5$ AT/Wb $\Rightarrow \mathcal{R}_{eff} = 2.55\times10^5$

$\Rightarrow \phi = \frac{600\times0.25}{5.24\times10^5} = 5.88\times10^{-4}$

① $A_1 = 150\,cm^2$ $A_2 = 100\,cm^2$ $l_1 = 90\,cm$ $l_2 = 30\,cm$

$$R_1 = \frac{130\times10^{-2}}{(4\pi\times10^{-7})(2500)(150\times10^{-4})} = \frac{13\times10^{1}}{15\pi\times10^{-6}} = 2.75\times10^{4}\ AT/wb$$

$$R_2 = \frac{45\times10^{-2}}{(4\pi\times10^{-7})(2500)(100\times10^{-4})} = \frac{4.5\times10^{-1}}{\pi\times10^{-5}} = 1.43\times10^{4}\ AT/wb$$

$$\mathcal{E} = NI = 200\times 1 = 200 \Rightarrow \phi = \frac{2\times10^{2}}{4.18\times10^{4}} = 0.48\times10^{-2} = 4.78\,mWb$$

② $l = 8\times10^{-1}m$ $A = \frac{9\pi}{4}\times10^{-4}m^2$ $N = 6\times10^{2}$

(a) $\phi = 5\times10^{-4}\,wb$ $R = \frac{8\times10^{-1}\times4}{4\pi\times10^{-7}\times9\pi\times10^{-4}} = \frac{8}{9\pi^2}\times10^{10}$

$$NI = \phi R \Rightarrow 6\times10^{2}\times I = 5\times10^{-4}\times\frac{8}{9\pi^2}\times10^{10} \qquad I = \frac{2}{27\pi^2}\times10^{5}\,A = 750.53A$$

(b) $R = \frac{8}{9\pi}\times\frac{10^{10}}{3\times10^{3}} = \frac{8}{27\pi^2}\times10^{7} \Rightarrow 6\times10^{2} I = 5\times10^{-4}\times\frac{8}{27\pi^2}\times10^{7}$

$$\Rightarrow I = \frac{400}{162\pi^2} = 0.25A$$

③ $l_m = 4\times10^{-1}m$ $A = 12\times10^{-4}m^2$ $l_{gap} = 5\times10^{-4}m$ $N = 400$

(a) $\phi = 168\times10^{-5}\,wb$ $B = 1.4T \Rightarrow H_{core} = 500\,A/m$

$$\mathcal{E}_{core} = H_{core}\,l_m = 500\times4\times10^{-1} = 200\,A\text{-turns}$$

$$H_{gap} = 1.4/\mu_0 = 7/2\pi\times10^{6}\,A/m \Rightarrow \mathcal{E}_{gap} = 7/2\pi\times10^{6}\times5\times10^{-4} = 557\,A\ turns$$

$$\Rightarrow \mathcal{E}_{total} = 757\,A\,turns = NI \Rightarrow I = 1.8925A$$

(b) $\mu_r = \frac{1.4}{4\pi\times10^{-7}\times500} = 2228.17$

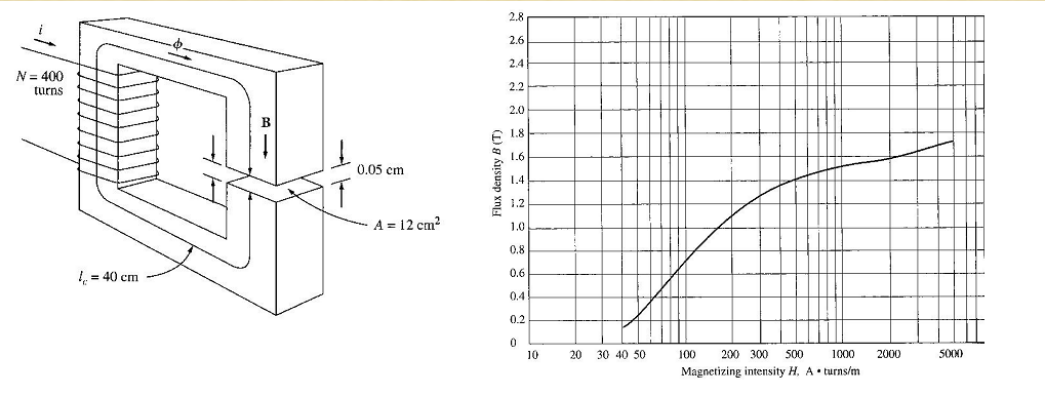
(c) $R_{coil} = 4\times10^{-1}/1.4 \times 500/12\times10^{-4} = 1.19\times10^{5}\ AT/wb$

3. A square magnetic core has a mean path length of 40 cm and a cross sectional area of 12 cm^2. There is a small gap of 0.05 cm in the structure of the otherwise whole core as shown in below figure. A 400 turn coil of wire is wrapped around one leg of the core. Use the B-H curve shown below and neglect magnetic leakage and fringing. Find:

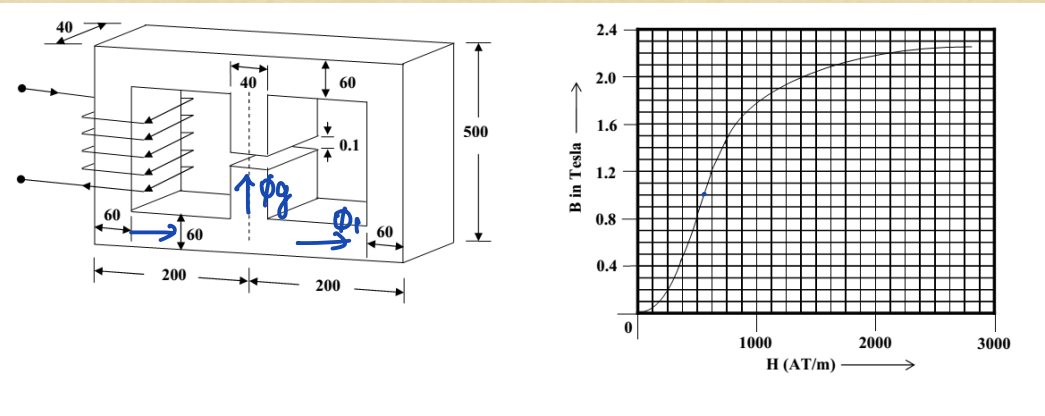
   (a) How much current is required to produce 1.68 mWb of flux in the air gap?

   (b) What is the core's relative permeability at that current level?

   (c) What is its reluctance?



4. In the magnetic circuit detailed in Figure with all dimensions in mm, calculate the required current to be passed in the coil having 200 turns in order to establish a flux of 1.28 mWb in the air gap. Neglect fringing effect and leakage flux. The B-H curve of the material is given below. Permeability of air may be taken as, $\mu_o = 4\pi$ x 10^{-7} H/m.



5. The magnetic circuit shown in below figure has two windings and two air gaps. The core can be assumed to be of infinite permeability. The core dimensions are indicated in the figure.

(4)

$B_g = \frac{1.28\times10^{-3}}{16\times10^{-4}} = 0.8T \Rightarrow H_g = B_g/\mu_o = 63.66\times10^4 AT/m \Rightarrow e_g = H_g l_g = 63.66AT$

$B_2 = 0.8T \Rightarrow H_2 = 500AT/m \Rightarrow e_2 = H_2 l_2 = 220AT \Rightarrow e_{tot} = 283.66AT$

$(l_m)_1 = 170+440+170 = 780mm \Rightarrow H_1 = \frac{283.66}{0.78} = 363.67 AT/m \Rightarrow B = 0.5T$

$\Rightarrow \phi_1 = B_1 A_1 = 1.2 mWb \Rightarrow \phi_T = 2.48 mWb$; $B_3 = 1.033 \Rightarrow H = 562.5 AT/m$

$\Rightarrow e = 438.8 AT$

$\Rightarrow e_{total} = 438.8 + 283.67 = 722.42AT \Rightarrow I = 722.42/200 = 3.61$

(5) $\mu_r = 0 \Rightarrow e_{core} = 0 \Rightarrow$ mmf only drops across gaps

$\Downarrow$

$R_{core} = 0$

$e_{g_1} = N_1 I_1$ $\qquad$ $e_{g_2} = N_1 I_1 + N_2 I_2$

$\frac{B_{g_1} g_1}{\mu_o} = N_1 I_1$ $\qquad$ $\frac{B_{g_2} g_2}{\mu_o} = N_1 I_1 + N_2 I_2$

$\Rightarrow B_{g_1} = \frac{\mu_o}{g_1} N_1 I_1$ $\qquad$ $B_{g_2} = \frac{\mu_o}{g_2}(N_1 I_1 + N_2 I_2)$

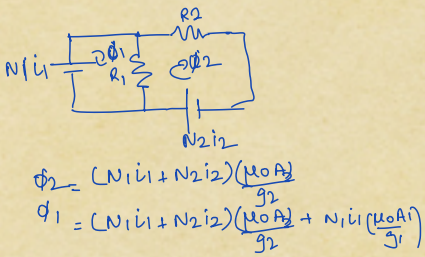
$\Rightarrow \phi_1 = B_{g_1} A_1 = \frac{\mu_o A_1 N_1 I_1}{g_1}$ $\qquad$ $\phi_2 = \frac{\mu_o A_2}{g_2}(N_1 I_1 + N_2 I_2)$

linked flux 1 (λ_1): $N_1(\phi_1+\phi_2) = N_1\mu_o\left[\frac{A_1}{g_1}(N_1 I_1) + \frac{A_2}{g_2}(N_1 I_1 + N_2 I_2)\right]$

linked flux 2 (λ_2): $N_2\phi_2 = \frac{N_2\mu_o A_2}{g_2}(N_1 I_1 + N_2 I_2)$

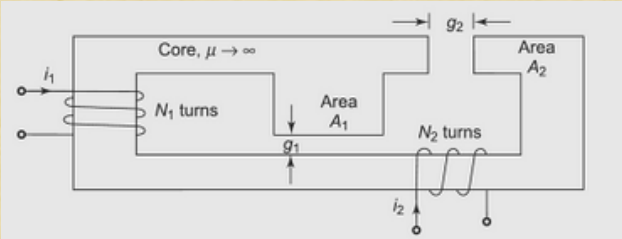
$\lambda_1 = L_{11} I_1 + L_{12} I_2$ $\qquad$ $\lambda_2 = L_{21} I_1 + L_{22} I_2$

$L_{11}, L_{22} \rightarrow$ self inductances ; $L_{12} = L_{21} \rightarrow$ mutual inductance



$\phi_2 = \frac{(N_1 i_1 + N_2 i_2)(\mu_0 A_2)}{g_2}$

$\phi_1 = (N_1 i_1 + N_2 i_2)\frac{(\mu_0 A_2)}{g_2} + N_1 i_1 \frac{(\mu_0 A_1)}{g_1}$



(a) Assuming coil 1 to be carrying a current I_1 and the current in coil 2 to be zero, calculate

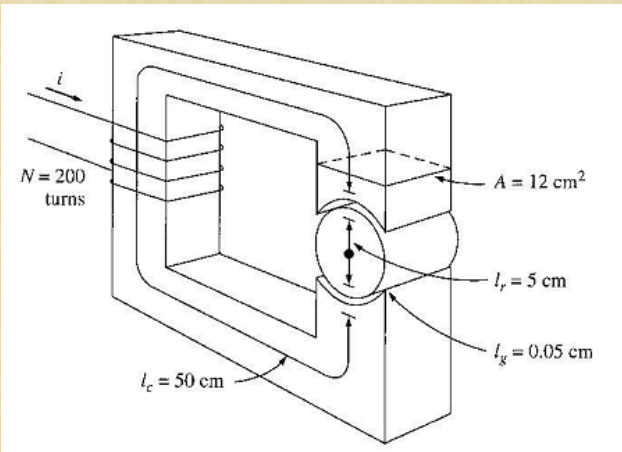
(i) The magnetic flux density in each of the air gaps

(ii) The flux linkage of winding 1, and

(iii) The flux linkage of winding 2.

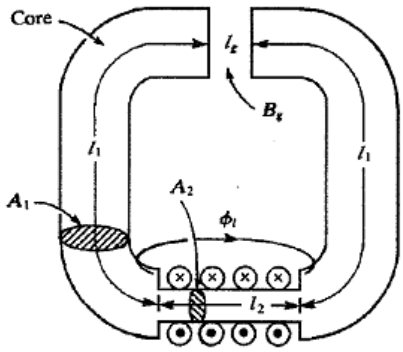
(b) Repeat part (a), assuming the current in winding 1 to be I_1 and winding 2 to be I_2

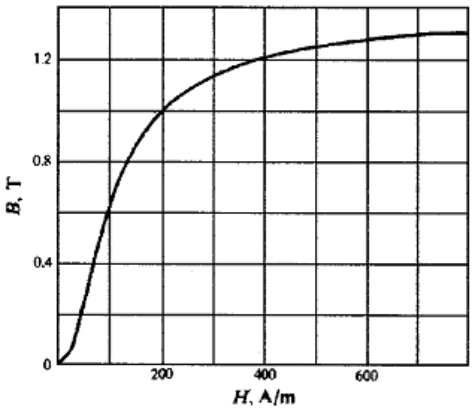
6. Figure shows a simplified rotor and stator for a dc motor. The mean path length of the stator is 50 cm, and its cross sectional area is 12 cm^2. The mean path length of the rotor is 5 cm, and its cross sectional area also may be assumed to be 12 cm^2. Each air gap between the rotor and the stator is 0.05 cm wide, and the cross sectional area of each air gap (including fringing) is 14 cm^2. The iron of the core has a relative permeability of 2000, and there are 200 turns of wire on the core. If the current in the wire is adjusted to be 1 A, what will the resulting flux density in the air gaps be?



7. Figure below shows a composite magnetic circuit of varying cross sectional area. The iron portion has the B-H characteristics as shown. Given N=400 turns, $l_1 = 4l_2$=40 cm, $A_1 = 2A_2$=10 cm^2, l_g = 2 mm, leakage flux, ϕ_l = 0.01 mWb. Calculate current required to establish an air gap flux density of 0.6 T.

leakage flux adds to iron flux & not airgap flux





(6) $R_{stator} = \frac{5\times10^{-1}}{4\pi\times10^{-7}\times2\times10^{3}\times12\times10^{-4}} = 1.66\times10^{5}$

$R_{rotor} = \frac{5\times10^{-2}}{4\pi\times10^{-7}\times2\times10^{3}\times12\times10^{-4}} = 1.66\times10^{4}$

$R_{air\ gap} = \frac{2\times5\times10^{-4}}{4\pi\times10^{-7}\times14\times10^{-4}} = 5.68\times10^{5}$

$R_{total} = 7.51\times10^{5}$ $\quad \phi = \frac{200}{7.51\times10^{5}} = 2.66\times10^{-4}$ wb

$B = \frac{\phi}{A_g} = \frac{2.66\times10^{-4}}{14\times10^{-4}} = 0.19\,T$

(7) $B_a = 0.6\,T \Rightarrow \phi_g = 6\times10^{-1}\times10^{-3} = 6\times10^{-4}$ wb

$\Rightarrow \phi_{total} = 6.1\times10^{-4}$ wb $\Rightarrow (B_{Fe})_1 = \frac{6.1\times10^{-4}}{10^{-3}} = 0.61\,T$

$(B_{Fe})_2 = \frac{6.1\times10^{-4}}{5\times10^{-4}} = 1.22\,T$

$(H_{Fe})_1 = 100$ A/m $\quad (H_{Fe})_2 = 400$ A/m

$400\times10\times10^{-2}$

$e_{Fe} = (H_{Fe})_1 l_1 + (H_{Fe})_2 l_2 = 40 + 40 = 80$ AT

$H_g = \frac{0.6}{4\pi\times10^{-7}} = 4.77\times10^{5}$ A/m $\Rightarrow e_g = 4.77\times10^{5}\times2\times10^{-3} = 954$ AT

$\Rightarrow e_{total} = 1074$ AT $= NI \Rightarrow I = 8.69$ A

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Tutorial 2

1. A 100 kVA, 1100/230 V, 50 Hz transformer has a HV winding resistance of 0.1 Ω and a leakage reactance of 0.4 Ω. The LV winding has a resistance of 0.006 Ω and a leakage reactance of 0.01 Ω. Find the equivalent winding resistance, reactance and impedance referred to the HV and LV sides.

2. A 230/6 V, single-phase transformer is tested with its secondary winding short circuited. A low voltage (20 V) is applied to the primary terminals and it then takes an input current of 1 A; the power supplied being 10 W. Calculate the values of leakage reactance and resistance, referred to the primary side and then to the secondary side. If the magnetising impedance is neglected, calculate the secondary terminal voltage when a load impedance of value (0.12+j0.09) Ω is connected.

3. A 50 kVA, 2400/240 V, 60 Hz distribution transformer has a leakage impedance of (0.72+j0.92) Ω in the high-voltage winding and (0.007+j0.009) Ω in the low-voltage winding. At rated voltage and frequency, the impedance Z_ϕ of the shunt branch (equal to the impedance of R_c and jX_m in parallel) accounting for the exciting circuit is (632+j43.7) Ω when viewed from the low-voltage side.
   The transformer is used to step down the voltage at the load end of a feeder whose impedance is (0.3+j1.6) Ω. The voltage V_s at the sending end of the feeder is 2400 V. Find the voltage at the secondary terminals of the transformer when the load connected to its secondary draws rated current from the transformer and the power factor of the load is 0.8 lagging. Neglect the voltage drops in the transformer and feeder caused by the exciting current.

4. A single-phase load is supplied through a 35 kV feeder whose impedance is (95+j360) Ω and a 35 kV/2400 V transformer whose equivalent impedance is (0.23+j1.27) Ω referred to its low-voltage side. The load is 160 kW at 0.89 leading power factor and 2340 V.

   (a) Compute the voltage at the high-voltage terminals of the transformer.

   (b) Compute the voltage at the sending end of the feeder.

   (c) Compute the power and reactive power input at the sending end of the feeder.

5. A 50 kVA, 2200/110 V transformer when tested gave the following results:
   OC test: measurement on the LV side: 400 W, 10 A, 110 V
   SC test: measurement on the HV side: 808 W, 20.5 A, 90 V
   Compute all the parameters of the equivalent circuit referred to the

   (a) HV side of the transformer

   (b) LV side of the transformer

1. A 100 kVA, 1100/230 V, 50 Hz transformer has a HV winding resistance of 0.1 Ω and a leakage reactance of 0.4 Ω. The LV winding has a resistance of 0.006 Ω and a leakage reactance of 0.01 Ω. Find the equivalent winding resistance, reactance and impedance referred to the HV and LV sides.

a) HV side: $a = \frac{1100}{230} = 4.78$

$r_2' = a^2 r_2 = 0.006 \times 4.78^2 = 0.137\,\Omega$ $\quad R_1 = r_1 + r_2' = 0.237\,\Omega$

$x_2' = a^2 x_2 = 0.01 \times 4.78^2 = 0.228\,\Omega$ $\quad X_1 = x_1 + x_2' = 0.628\,\Omega$

b) LV side:

$r_1' = r_1/a^2 = 0.0044\,\Omega$ $\quad R_2 = r_1' + r_2 = 0.0104\,\Omega$

$x_1' = x_1/a^2 = 0.0175\,\Omega$ $\quad X_2 = x_1' + x_2 = 0.0275\,\Omega$

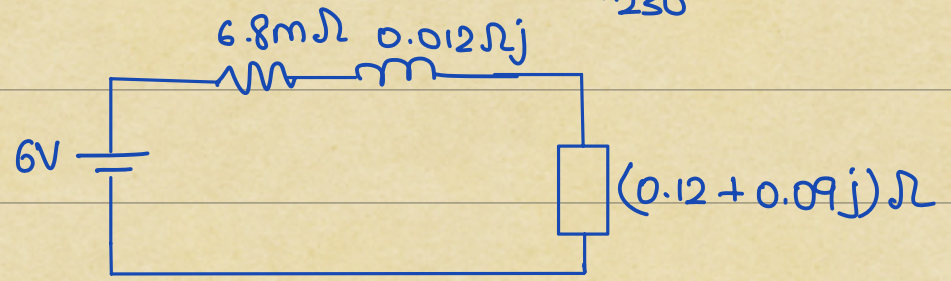
2. A 230/6 V, single-phase transformer is tested with its secondary winding short circuited. A low voltage (20 V) is applied to the primary terminals and it then takes an input current of 1 A; the power supplied being 10 W. Calculate the values of leakage reactance and resistance, referred to the primary side and then to the secondary side. If the magnetising impedance is neglected, calculate the secondary terminal voltage when a load impedance of value (0.12+j0.09) Ω is connected.

$R = \frac{P}{I^2} = 10\,\Omega$ (1° side)

$Z = \frac{V}{I} = 20\,\Omega \Rightarrow X = 10\sqrt{3}\,\Omega$ (1° side)

R (2° side) $= 10 \times \left(\frac{6}{230}\right)^2 = 6.8\ m\Omega$

X (2° side) $= 10\sqrt{3} \times \left(\frac{6}{230}\right)^2 = 0.012\,\Omega$



$V_{load} = 6 \times \frac{0.12+0.09j}{(6.8\times10^{-3}+0.012j)+(0.12+0.09j)}$

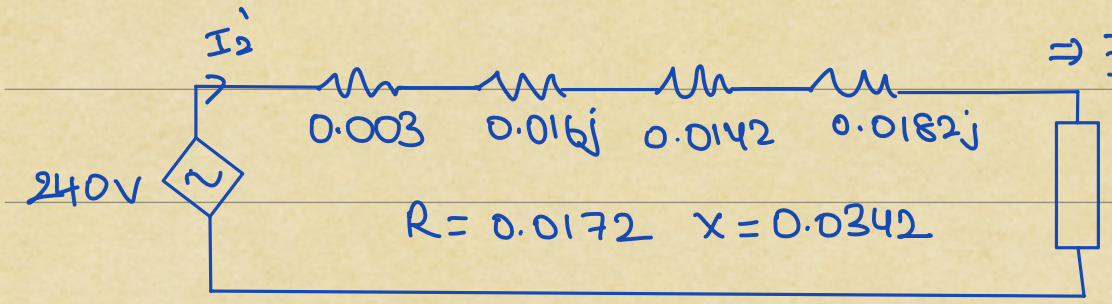
$= 5.5346\ \angle -1.8878$

3. A 50 kVA, 2400/240 V, 60 Hz distribution transformer has a leakage impedance of (0.72+j0.92) Ω in the high-voltage winding and (0.007+j0.009) Ω in the low-voltage winding. At rated voltage and frequency, the impedance Z_ϕ of the shunt branch (equal to the impedance of R_c and jX_m in parallel) accounting for the exciting circuit is (632+j43.7) Ω when viewed from the low-voltage side.
The transformer is used to step down the voltage at the load end of a feeder whose impedance is (0.3+j1.6) Ω. The voltage V_s at the sending end of the feeder is 2400 V. Find the voltage at the secondary terminals of the transformer when the load connected to its secondary draws rated current from the transformer and the power factor of the load is 0.8 lagging. Neglect the voltage drops in the transformer and feeder caused by the exciting current.

$a = 10$

$I_2' = \frac{50 \times 10^3}{240} = 208.33$ A

$\Rightarrow \overline{I_2'} = 208.33 \angle -36.86^\circ$

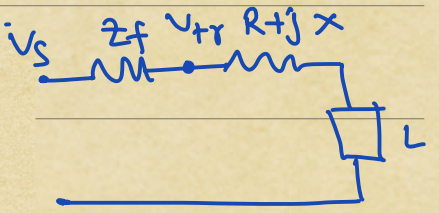


$|V_L| + |I_2'|(R\cos\phi + X\sin\phi) = |V_1|$

$\Rightarrow |V_L| = 240 - 208.33(0.0172 \times 0.8 + 0.0342 \times 0.6)$

$= 232.86$ V

4. A single-phase load is supplied through a 35 kV feeder whose impedance is (95+j360) Ω and a 35 kV/2400 V transformer whose equivalent impedance is (0.23+j1.27) Ω referred to its low-voltage side. The load is 160 kW at 0.89 leading power factor and 2340 V.



(a) Compute the voltage at the high-voltage terminals of the transformer.

(b) Compute the voltage at the sending end of the feeder.

(c) Compute the power and reactive power input at the sending end of the feeder.

a) $Z_{feeder} = (95+360j) \times \left(\frac{2400}{35000}\right)^2 = (0.4467 + 1.6927j)\ \Omega$

$160 \times 10^3 = V_2' I_2' \cos\phi \Rightarrow |I_2'| = 76.827$

$\Rightarrow \overline{I_2'} = 76.827 \cos^{-1}(0.89) = 76.827 \angle 27.1268$

$|\text{Voltage}| = |2340 + 76.827\angle 27.1268(0.23+1.27j)| = 2313.19$V

Voltage at HV terminals $= \frac{35000}{2400} \times 2313.19 = 33.73$ kV

b) $V_S = V_{tr} + \overline{I_2'}(Z_f) = 2313.1856 \angle 2.35 + 76.827 \angle 27.1267 (0.4467 + 1.6927j)$

$= 2293.6732 \angle 5.6606$

$\Rightarrow$ actual voltage @ sending end $2293 \times \frac{35000}{2400} = 33439.3$

c) $P = \underline{V}\underline{I}_s^* = 163992.6741 - 64486.3738j$

real complex

5. A 50 kVA, 2200/110 V transformer when tested gave the following results:
   OC test: measurement on the LV side: 400 W, 10 A, 110 V
   SC test: measurement on the HV side: 808 W, 20.5 A, 90 V
   Compute all the parameters of the equivalent circuit referred to the

   (a) HV side of the transformer

   (b) LV side of the transformer

$a = 20$

$r_c = \frac{V_{oc}^2}{P_{oc}} = 30.25\,\Omega \qquad i_c = \frac{V_{oc}}{r_c} = 3.6364\,A$

$i_{xm} = \sqrt{10^2 - 3.6364^2} = 9.3154\,A \Rightarrow x_m = \frac{V_{oc}}{i_{xm}} = 11.8082\,\Omega$ } LV side

$r_c = 30.25 \times 20^2 = 12100\,\Omega \qquad x_m = 11.8082 \times 20^2 = 4723.28\,\Omega \rightarrow$ HV side

$R = \frac{P_{sc}}{I_{sc}^2} = \frac{808}{20.5^2} = 1.922\,\Omega \rightarrow$ HV side

$R' = 1.922 \times \frac{1}{400} = 0.048\,\Omega \rightarrow$ LV side

$Z_{sc} = \frac{V_{sc}}{I_{sc}} = \frac{90}{20.5} = 4.3902\,\Omega$

$X = \sqrt{Z_{sc}^2 - R^2} = 3.9472\,\Omega \rightarrow$ HV side $\qquad 0.0099\,\Omega \rightarrow$ LV side

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Tutorial 3

1. A 200 V/400 V, 1-phase, 50 Hz, 4 kVA transformer has the following equivalent circuit parameters referred to the low-voltage side: $R = 0.15\,\Omega$, $X = 0.37\,\Omega$, $R_m = 600\,\Omega$, $X_m = 300\,\Omega$. For a high voltage current of 10 A at a lagging power factor of 0.8, calculate (a) the low-voltage input current (b) the efficiency and (c) the voltage at the terminals of the high voltage side.

2. In a 25 kVA, 2000 V/200 V, 1-phase transformer, the iron and full-load copper losses are equal to 350 and 400 W respectively.

   (a) Calculate the efficiency when the unit supplies at UPF

   (i) full load

   (ii) half the full load

   (b) Determine the load for maximum efficiency and the iron and copper losses in this case.

3. The resistances and leakage reactances of a 10 kVA, 50 Hz, 2200 V/220 V distribution transformer are as follows: $R_1 = 4\,\Omega$, $R_2 = 0.04\,\Omega$, $X_1 = 5\,\Omega$ and $X_2 = 0.005\,\Omega$

   (a) Find the total leakage impedance referred to (i) HV side (ii) LV side.

   (b) If the transformer is supplied from an AC source of 2200 V at the primary side, and a full load current at 0.8 pf lag flows in the secondary, find the terminal voltage across the secondary.

   (c) Find the same as in part (b) for a full-load pf of 0.8 leading.

   (d) Considering the core losses to be 80 W, find the efficiency under condition given in part (b) and part (c).

   (e) If the load in part (b) gets short-circuited, find the steady-state current in the HV lines, assuming that the voltage applied to the transformer remains unchanged.

4. The nameplate on a 50 MVA, 60 Hz single-phase transformer indicates that it has a voltage rating of 8 kV/78 kV. An open-circuit test is conducted from the low-voltage side, and the corresponding instrument readings are 8 kV, 62.1 A, and 206 kW. Similarly, a short-circuit test from the low-voltage side gives readings of 674 V, 6.25 kA, and 187 kW.

   (a) Calculate the equivalent circuit parameters of the trans- former as referred to the low-voltage terminals.

   (b) Calculate the equivalent circuit parameters as referred to the high- voltage terminals.

   (c) Determine the efficiency and voltage regulation if the transformer is operating at the rated voltage and load (unity power factor).

   (d) Determine the load at UPF for which the efficiency is maximum.

1. A 200 V/400 V, 1-phase, 50 Hz, 4 kVA transformer has the following equivalent circuit parameters referred to the low-voltage side: $R = 0.15\,\Omega$, $X = 0.37\,\Omega$, $R_m = 600\,\Omega$, $X_m = 300\,\Omega$. For a high voltage current of 10 A at a lagging power factor of 0.8, calculate (a) the low-voltage input current (b) the efficiency and (c) the voltage at the terminals of the high voltage side.

$a = \frac{1}{2}$

$I_2' = 10 \times \frac{400}{200} = 20\,A \Rightarrow \bar{I}_2' = 20\angle -36.87^\circ$ (refered to LV)

$V_2' = 200\,V$ (referred to LV side)

ΔV (impedance) $= (0.15 + 0.37j)(\bar{I}_2') = 6.84 + 4.12j$

$V_1 = V_2' + \Delta V = 206.84 + 4.12j \Rightarrow |V_1| = 206.9$

$I_C = \frac{V_1}{R_m} = 0.345\,A \quad I_m = \frac{V_1}{x_m} = 0.69\,A$

$I_o = I_c - jI_m = 0.345 - 0.69j$

$I_1 = I_o + I_2' = 16.345 - 12.69j \Rightarrow |I_1| = 20.6A$

$P_{out} = 200 \times 10 \times 0.8 = 3200W$

$P_{Cu\ loss} = (I_2')^2 R = 400 \times 0.15 = 60W$

$P_{core\ loss} = \frac{V_1^2}{R_m} = \frac{(206.9)^2}{600} = 71W \Rightarrow P_{in} = 3200 + 60 + 71 = 3331W$

$\eta = \frac{3200}{3331} \times 100 = 96.1\%$

HV terminal voltage $V_2 = aV_1 = 2 \times 206.9 = 413.8V$

2. In a 25 kVA, 2000 V/200 V, 1-phase transformer, the iron and full-load copper losses are equal to 350 and 400 W respectively.

   (a) Calculate the efficiency when the unit supplies at UPF
      (i) full load
      (ii) half the full load
   (b) Determine the load for maximum efficiency and the iron and copper losses in this case.

(a)

(i) $P_{out} = 25000\,W \quad P_{in} = 25000 + 350 + 400 = 25750$

$$\Rightarrow \eta = \frac{25000}{25750} \times 100 = 97.087\%$$

(ii) half load:

$P_{out} = 25000 \times 0.5 = 12500 \qquad P_{fe} = 350\omega$

$P_{cu} = 0.25 \times 400 = 100\omega \Rightarrow P_{in} = 12950\omega$

$$\Rightarrow \eta = \frac{12500}{12950} \times 100 = 96.525\%$$

(b) $P_{cu} = P_{fe} \Rightarrow x^2 \times 400 = 350 \Rightarrow x = 0.9354$ 93.5% of full load

$P_{cu} = P_{fe} = 350\omega \qquad P_{out} = 0.935 \times 25000 = 23385.25$

$$\eta = \frac{23385.25}{24085.25} \times 100 = 97.094\%$$

3. The resistances and leakage reactances of a 10 kVA, 50 Hz, 2200 V/220 V distribution transformer are as follows: $R_1 = 4\,\Omega$, $R_2 = 0.04\,\Omega$, $X_1 = 5\,\Omega$ and $X_2 = 0.005\,\Omega$

(a) Find the total leakage impedance referred to (i) HV side (ii) LV side.

(b) If the transformer is supplied from an AC source of 2200 V at the primary side, and a full load current at 0.8 pf lag flows in the secondary, find the terminal voltage across the secondary.

(c) Find the same as in part (b) for a full-load pf of 0.8 leading.

(d) Considering the core losses to be 80 W, find the efficiency under condition given in part (b) and part (c).

(e) If the load in part (b) gets short-circuited, find the steady-state current in the HV lines, assuming that the voltage applied to the transformer remains unchanged.

$a = 10$

$\%\ \text{regulation} = \left(1 - \frac{\text{value}}{\text{rate}}\right) 100$

(a) (i) HV side: $R_H = 4 + 100 \times 0.04 = 8 \qquad X_H = 5 + 100 \times 0.005 = 5.5$

$\Rightarrow Z_H = 8 + 5.5j$

(ii) LV side: $Z_L = 0.08 + 0.055j$

(b) $I_2 = \frac{10000}{220} = 45.45A \qquad \phi = -36.87 \Rightarrow I_2 = 45.45\angle -36.87$

$= 36.36 - 27.27j$

$\Delta V = (I_2)(Z_L) = 4.41 - 0.18j \Rightarrow V_2 = 220 - \Delta V = 215.6 - 0.18j$

$|V_2| = 215.6V$

(c) $I_2 = 36.36 - 27.27j \Rightarrow \Delta V = (I_2)(Z_c) = 1.41 + 4.18j$

$\Rightarrow V_2 = 220 - \Delta V = 218.6 - 4.18j \Rightarrow |V_2| = 219.3V$

(d) $P_{cu} = (45.45)^2(0.08) = 165.3$ Loss = 245.3Ω

(b) $P_{out} = 215.6 \times 45.45 \times 0.8 = 7838.4\Omega$ $\eta = 96.96\%$.

(c) $P_{out} = 219.3 \times 45.45 \times 0.8 = 7973.7\Omega$ $\eta = 97.02\%$.

(e) $|Z_H| = \sqrt{8^2+5.5^2} = 9.7\Omega$ $I_{SC} = \frac{2200}{9.7} = 227A$

4. The nameplate on a 50 MVA, 60 Hz single-phase transformer indicates that it has a voltage rating of 8 kV/78 kV. An open-circuit test is conducted from the low-voltage side, and the corresponding instrument readings are 8 kV, 62.1 A, and 206 kW. Similarly, a short-circuit test from the low-voltage side gives readings of 674 V, 6.25 kA, and 187 kW.

a = 9.75

(a) Calculate the equivalent circuit parameters of the trans- former as referred to the low-voltage terminals.

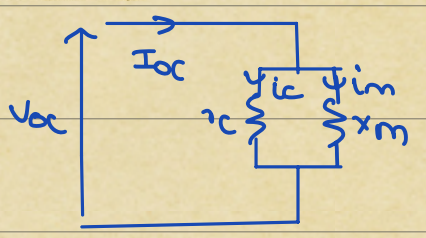
(b) Calculate the equivalent circuit parameters as referred to the high- voltage terminals.

(c) Determine the efficiency and voltage regulation if the transformer is operating at the rated voltage and load (unity power factor).

(d) Determine the load at UPF for which the efficiency is maximum.

a) oc test: $V_{OC} = 8KV$ $I_{oc} = 62.1A$ $P_{oc} = 206K\omega$

$206 \times 10^3 = \frac{(8 \times 10^3)^2}{r_c} \Rightarrow r_c = 310.7\Omega$

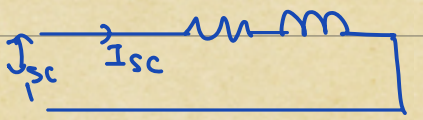


$I_c = \frac{8 \times 10^3}{310.7} = 25.75A$

$I_m = \sqrt{I_{oc}^2 - I_c^2} = 56.51A \Rightarrow X_m = \frac{8 \times 10^3}{56.51} = 141.56\Omega$

sc test:

$R = \frac{P_{SC}}{I_{SC}^2} = 4.7872m\Omega$

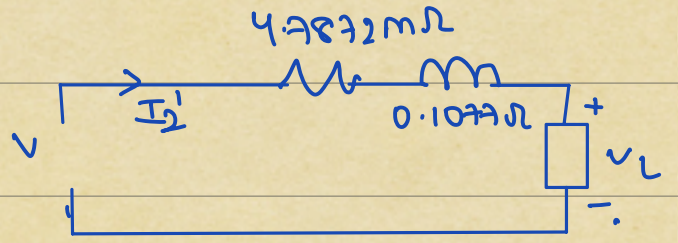


$Z = \frac{V_{SC}}{I_{SC}} = 0.1076\Omega \Rightarrow X = \sqrt{Z^2 - R^2} = 0.1077$

b) values $\times \left(\frac{78}{8}\right)^2$

c)

$I_2' = \frac{5\times10^7}{8000} = 6250\,A$



$V_{load} = V - I_2'(R)$

$= 8000 - 6250\times 4.7872\times10^{-3} = 7970.8\,V$

% regulation $= \left(1 - \frac{7970.8}{8000}\right)100 = 0.374\%$

$P_{out} = 7970.8\times 6250 = 49.8175\,MW$

Fe loss $= 206\,kW$ $\Rightarrow \eta = 99.217\%$

Cu loss $= (6250)^2\times(4.7872\times10^{-3}) = 187\,kW$

d) $n^2\times 187 = 206 \Rightarrow$ load factor for max $\eta = 1.05$

5. The following data were obtained for a 20 kVA, 50 Hz, 2400 V/240 V distribution transformer when tested:
With HV winding open circuited: 240 V, 1.006 A, 126.6 W — Fe loss
With LV side short circuited: 57.5 V, 8.34 A, 284 W — Cu loss

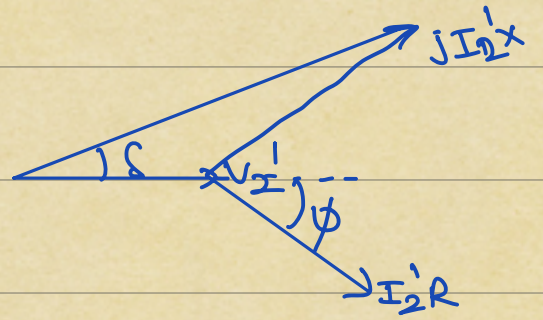
(a) Compute the efficiency at full load current and the rated terminal voltage at 0.8 power factor lagging.

(b) Assume that the load power factor is varied while the load current and secondary terminal voltages are held constant. Use phasor diagram to determine the load power factor for which the regulation is highest. What is this regulation?

(a) $P = 20\times10^3\times0.8 = 16000\,W \Rightarrow \eta = \frac{16000}{16410.6} = 97.49\%$

(b) $R = \frac{P_{sc}}{I_{sc}^2} = 4.083\,\Omega$

$Z = \frac{V_{cc}}{I_{sc}} = 6.8945\,\Omega \Rightarrow X = 5.5554\,\Omega$



$\delta = 0$

$2400 + 8.34\angle\phi\,(4.083 + j\,5.55)$

$\Rightarrow \phi = -53.659°$

power factor $= 0.59$ lagging 2% regulation

5. The following data were obtained for a 20 kVA, 50 Hz, 2400 V/240 V distribution transformer when tested:
   With HV winding open circuited: 240 V, 1.006 A, 126.6 W
   With LV side short circuited: 57.5 V, 8.34 A, 284 W

   (a) Compute the efficiency at full load current and the rated terminal voltage at 0.8 power factor lagging.

   (b) Assume that the load power factor is varied while the load current and secondary terminal voltages are held constant. Use phasor diagram to determine the load power factor for which the regulation is highest. What is this regulation?

EE2006: Electrical Machines

Tutorial 4

1. A separately excited DC motor has back-emf constant, kϕ = 1.44 V/rad/s, Ra = 0.86 Ω. When it is operating at 150 rad/s, the armature current is Ia = 40 A. The terminal voltage Vt is held constant under all conditions.
(a) Compute Vt
(b) Compute the no-load speed in rad./s (Neglect core and mechanical losses)
[Ans: (a) 250.4 V (b) 173.89 rad/s]

2. A 200 V DC shunt motor has an armature resistance of 0.4 Ω. When operated from the rated supply voltage at full load, the motor runs at 500 rpm and draws an armature current of 25 A. If a resistance of 1.6 Ω is placed in series with the armature, find the speed at (i) full load torque (ii) three times full load torque. Neglect core and mechanical losses.
[Ans: (i) 394.736 rpm (ii) 131.58 rpm]

$K\phi I_a = 100 \quad (I_a)_{FL} = 69.44$

3. A separately excited DC motor is rated for a full load torque of 100 Nm. For an applied voltage of 240 V, the no-load and full load speeds of the motor are 1595 rpm and 1195 rpm respectively, find
(a) (i) The back-emf constant ke of the machine
(ii) The armature resistance Ra
(b) If the applied voltage and the field of the machine are reduced to 83.33% and 66.66% of the previous values respectively, find
(i) The corresponding no load speed
(ii) The corresponding full load speed
[Ans: (a-i) 1.4369 V/rad/s (a-ii) 0.8648 Ω (b-i) 1993.76 rpm (b-ii) 1093.79 rpm]

162 rps 125.1

$K\phi = 1.44$

$240 = 1.44(125.1) + 69.44(R_a) \quad R_a = 0.861$

$200 = 0.96(\omega) + 104.17(0.861)$

4. A 10 kW, 250 V, 1200 rpm DC shunt motor has a full load efficiency of 80%. The field and armature resistances are 125 Ω and 0.2 Ω respectively. The speed of the motor is to be reduced to 75% with load torque remaining constant. What resistance should be inserted in the armature circuit. Neglect stray losses.
[Ans: 1.252 Ω]

Output $\Rightarrow$ I/P $= 12500W = 250(I_a + I_f) \Rightarrow I_a + I_f = 50$

$I_f = 250/125 = 2 \Rightarrow I_a = 48A$

$250 = 48 \times 0.2 + K.(125.66) \Rightarrow K = 1.91$

$250 = 1.91 \times 94.25 + R \times 48 \Rightarrow R = 1.46$

5. A 250 V DC shunt motor has an armature resistance of 0.2 Ω and shunt field resistance of 100 Ω. It takes 60 A when running at a speed of 600 rpm. If the speed is to be increased to 800 rpm, Find the resistance of the field regulator to be inserted. Assume the magnetisation characteristic to be a straight line and the load torque remains constant.
[Ans: 35.668 Ω]

$I_S = 2.5 \quad K_1 = 2.796$

$250 = 57.5 \times 0.2 + K(62.832)$

$I_{a2} \cdot K_2 = \text{const}$

6. A 250 V shunt motor develops a total torque of 105 Nm and takes 36 A at 750 rpm. The armature and shunt field resistances are 0.4 Ω and 250 Ω respectively. If the speed is to be increased to 1000 rpm, determine the percentage reduction of the field and additional resistance to be inserted in the field circuit. Total torque developed at 1000 rpm is 75 Nm. Assume the magnetisation characteristic to be a straight line.
[Ans: 24.77%, 82.3 Ω]

$K \propto I_S \propto 1/R_{Sh}$

$\phi \propto I_S$

$I_S = 1A \qquad \omega_2 = 104.72$

$K \cdot I_a = 105 \Rightarrow K = 3 \qquad 250 = 3(78.54) + (35)(0.4)$

$K_2 I_2 = 75 \qquad 250 = K_2(104.72) + \frac{75}{K_2}(0.4 + R)$

$\Rightarrow K_2 = 2.26 \quad I_2 = 33.185 \quad \frac{R_2}{R_1} = \frac{K_1}{K_2} \Rightarrow R_2 = 331.86$

Check ω/K_e if it is too much pack

$T = K I_f I_a$

$E = K I_f \omega$

$\%.\text{field} = \left|\frac{I_1 - I_2}{I_1}\right| \times 100 = \left|\frac{R_2 - R_1}{R_2}\right| = 24.66\%$

1. A separately excited DC motor has back-emf constant, $k\phi$ = 1.44 V/rad/s, Ra = 0.86 Ω. When it is operating at 150 rad/s, the armature current is Ia = 40 A. The terminal voltage Vt is held constant under all conditions.
(a) Compute Vt
(b) Compute the no-load speed in rad./s (Neglect core and mechanical losses)
[Ans: (a) 250.4 V (b) 173.89 rad/s]

Back emf $E = K_\phi \times \omega = 1.44 \times 1.50 = 216\,V$

(a) $V_t = E + I_a R_a = 250.4$

(b) No load $\Rightarrow I_a = 0 \Rightarrow V_t = K_\phi \omega_0 \Rightarrow \omega_0 = \frac{250.4}{1.44} = 173.89$

2. A 200 V DC shunt motor has an armature resistance of 0.4 Ω. When operated from the rated supply voltage at full load, the motor runs at 500 rpm and draws an armature current of 25 A. If a resistance of 1.6 Ω is placed in series with the armature, find the speed at (i) full load torque (ii) three times full load torque. Neglect core and mechanical losses.
[Ans: (i) 394.736 rpm (ii) 131.58 rpm]

$T \propto I$

$E_1 = 200 - (25 \times 0.4) = 190\,V$

$\frac{\omega_1}{\omega_2} = \frac{E_1}{E_2} = \frac{N_1}{N_2}$ (ω in rad/s, N in rpm)

(i) $R = 1.6 + 0.4 = 2\,\Omega \Rightarrow E_2 = 200 - (25 \times 2) = 150\,V$

$N_2 = \frac{150}{190} \times 500 = 394.736$ rpm

(ii) $E_3 = 200 - (75 \times 2) = 50\,V$

$\Rightarrow N_3 = \frac{50}{190} \times 500 = 131.58$ rpm

3. A separately excited DC motor is rated for a full load torque of 100 Nm. For an applied voltage of 240 V, the no-load and full load speeds of the motor are 1595 rpm and 1195 rpm respectively, find
(a) (i) The back-emf constant ke of the machine
(ii) The armature resistance Ra
(b) If the applied voltage and the field of the machine are reduced to 83.33% and 66.66% of the previous values respectively, find
(i) The corresponding no load speed
(ii) The corresponding full load speed
[Ans: (a-i) 1.4369 V/rad/s (a-ii) 0.8648 Ω (b-i) 1993.76 rpm (b-ii) 1093.79 rpm]

$\omega_0 = 167$ rad/s

$\omega_{FL} = 125.1$ rad/s

$T \propto K_e \phi I_a$

$\Rightarrow I_a \propto 1/K_e$

$K_e = \frac{240}{167} = 1.4369$ V/rad/s

(a) Full load: $T = K_e I_a \Rightarrow 100 = 1.4369 I_a \Rightarrow I_a = 69.6\,A$

$E_{FL} = K_e \omega_{FL} = 1.4369 \times 125.1 = 179.8\,V$

$240 = 179.8 + I_a R_a \Rightarrow R_a = 0.864\ \Omega$

(b) $V = 200V$ $\quad K_e \propto \phi \Rightarrow K_e = 0.958$

no load speed: $V/K_e = 208.747$ rad/s

$200 = 0.958\,\omega_{FL} + 104.4029 \times 0.864 \Rightarrow \omega_{FL} = 114.510$ rad/s

4. A 10 kW, 250 V, 1200 rpm DC shunt motor has a full load efficiency of 80%. The field and armature resistances are 125 Ω and 0.2 Ω respectively. The speed of the motor is to be reduced to 75% with load torque remaining constant. What resistance should be inserted in the armature circuit. Neglect stray losses.
[Ans: 1.252 Ω]

P_{out}

$P_{in} = 5/4 \times 10K = 12.5KW$

$I_L = \frac{P_{in}}{250} = 50A$

$I_{sh} = \frac{250}{125} = 2A \Rightarrow I_a = 48A$

$\Rightarrow E_{b1} = 250 - (48 \times 0.2) = 240.4V$

$\frac{E_{b2}}{E_{b1}} = \frac{3}{4} \Rightarrow E_{b2} = 180.3 = 250 - 48(0.2 + R) \Rightarrow R = 1.252\ \Omega$

5. A 250 V DC shunt motor has an armature resistance of 0.2 Ω and shunt field resistance of 100 Ω. It takes 60 A when running at a speed of 600 rpm. If the speed is to be increased to 800 rpm, Find the resistance of the field regulator to be inserted. Assume the magnetisation characteristic to be a straight line and the load torque remains constant.
[Ans: 35.668 Ω]

$I_{sh} = \frac{250}{100} = 2.5A \Rightarrow I_a = 57.5A$

$T \propto \phi I_a$

$T \propto I_f I_a$

$E_{b1} = 250 - (57.5 \times 0.2) = 238.5V$

$2.5 \times 57.5 = I_{f2} I_{a2} \Rightarrow I_{a2} = \frac{143.72}{I_{f2}}$

$\frac{E_{b2}}{E_{b1}} = \frac{\phi_2 N_2}{\phi_1 N_1} = \frac{I_{f2}}{I_{f1}} \cdot \frac{N_2}{N_1} \Rightarrow E_{b2} = 127.2 I_{f2}$

$127.2\,I_{f2} = 250 - \left(\frac{143.75}{I_{f2}}\right)(0.2) \Rightarrow I_{f2} = 1.843\,A$

$R_{sh} = \frac{250}{1.843} = 135.65 \Rightarrow 35.65\,\Omega$ extra

6. A 250 V shunt motor develops a total torque of 105 Nm and takes 36 A at 750 rpm. The armature and shunt field resistances are 0.4 Ω and 250 Ω respectively. If the speed is to be increased to 1000 rpm, determine the percentage reduction of the field and additional resistance to be inserted in the field circuit. Total torque developed at 1000 rpm is 75 Nm. Assume the magnetisation characteristic to be a straight line.
[Ans: 24.77%, 82.3 Ω]

$I_{Sh} = 1A \Rightarrow I_a = 35A \Rightarrow E_{b1} = 236V$

$T \propto \phi I_a \quad E_b \propto \phi N$

$\frac{T_2}{T_1} = \frac{I_{f2} I_{a2}}{I_{f1} I_{a1}} = \frac{75}{105} \Rightarrow I_{a2} = \frac{25}{I_{f2}}$

$\frac{E_{b2}}{E_{b1}} = \frac{I_{f2} N_2}{I_{f1} N_1} \Rightarrow E_{b2} = 314.6 I_{f2}$

$314.6 I_{f2} = 250 - \frac{10}{I_{f2}} \Rightarrow I_{f2} = 0.752\,A$

$\%\ \text{red}^n \text{ in field} = \frac{I_{f1} - I_{f2}}{I_{f1}} \times 100 = 24.8\%$

$R_{ext} + 250 = \frac{250}{0.752} \Rightarrow R_{ext} = 82.45\,\Omega$

EE2006: Electrical Machines

Tutorial 5

$200 = K \cdot 73.3 + 10 \Rightarrow K = 0.1296$ (20)

$200 = 3.1 \omega + 12$

$E = K I_a \omega$

$200 = 0.1296 \times 10 \omega + 5$

1. A 200 V DC series motor runs at 700 rpm when operating at its full load current of 20 A.

$I_s = I_a$

The motor resistance is 0.5 Ω and the magnetic circuit can be assumed unsaturated.

What will be the speed if (i) the load torque is increased by 44% (ii) the motor current is 10 A.

$I_a = 24$ $\quad$ $I_a = 10$

[Ans: (i) 577.19 rpm (ii) 1436.84 rpm]

$217 = K(38)62.832 + 38(0.6) \Rightarrow K = 0.0813$

2. A 220 V DC series motor draws full load line current of 38 A at the rated speed of 600 rpm.

The motor has armature resistance of 0.4 Ω and the series field resistance is 0.2 Ω.

The brush voltage drop irrespective of load is 3.0 V, find:

(a) The speed of the motor when the load current drops to 19 A. $217 = 0.0813(19)(\omega) + 19(0.6)$

(b) The speed on removal of load when the motor takes only 1 A from supply. $217 = 0.0813\omega + 0.6$

(c) The internal horse power developed in each of the above cases. $P = \frac{E_b I_a}{746}$

Neglect the effect of armature reaction and saturation.

[Ans: (a) 1270.44 rpm (b) 25406.38 rpm (c) 5.236 hp, 0.29 hp]

3. A 440 V series motor has a total series resistance (including field) of 0.8 Ω.

It takes 110 A to give the rated output at 800 rpm. $440 = 110 \times 0.8 + 110 \times K \times 83.776$

$K = 0.0382$

Determine the resistance to be added to obtain rated torque:

(i) At starting (ii) At 600 rpm. $440 = 110 \times R \Rightarrow R = 4$

$440 = 110 \times R + 110 \times 0.0382 \times 62.831 \Rightarrow R = 1.6$

Assume that the flux is proportional to the current.

[Ans: (i) 3.2 Ω (ii) 0.8 Ω]

4. A separately excited DC generator has no-load voltage of 120 V at a field current of 2 A,

when driven at 1500 rpm. Assuming that it is operating on the straight line portion of its

saturation curve, calculate: $K = 0.382$ $\quad$ $E_g = K I \omega$

(a) The generated voltage when the field current is increased to 2.5 A. 150 V

(b) The generated voltage when the speed is reduced to 1400 rpm and the field current is

increased to 2.84 A. $2.84 \times 0.382 \times 1400 \times \pi/30$

[Ans: (a) 150 V (b) 159.04 V]

1. A 200 V DC series motor runs at 700 rpm when operating at its full load current of 20 A.

The motor resistance is 0.5 Ω and the magnetic circuit can be assumed unsaturated. $T \propto \phi I_a \propto I_a^2$

What will be the speed if (i) the load torque is increased by 44% (ii) the motor current is 10 A.

[Ans: (i) 577.19 rpm (ii) 1436.84 rpm]

Back EMF initial $E_{b1} = V - I_{a1}R = 200 - (20 \times 0.5) = 190V$

(i) $T \propto I_a^2 \Rightarrow I_{a2} = 1.2 I_{a1} = 24A$

$E_{b2} = 200 - 24 \times 0.5 = 188V$

N_2 (new speed) $\propto \frac{E_{b2}}{I_{a2}}$

$\Rightarrow \frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}} \times \frac{I_{a1}}{I_{a2}} = \frac{188 \times 20}{190 \times 24}$

$\Rightarrow N_2 = 577.19$ rpm

(ii) $E_{b3} = 200 - (10 \times 0.5) = 195$

$\Rightarrow N_3 = 700 \times \frac{195}{190} \times \frac{20}{10} = 1436.84$ rpm

2. A 220 V DC series motor draws full load line current of 38 A at the rated speed of 600 rpm.

The motor has armature resistance of 0.4 Ω and the series field resistance is 0.2 Ω.

The brush voltage drop irrespective of load is 3.0 V, find:

(a) The speed of the motor when the load current drops to 19 A.

(b) The speed on removal of load when the motor takes only 1 A from supply.

(c) The internal horse power developed in each of the above cases.

Neglect the effect of armature reaction and saturation.

[Ans: (a) 1270.44 rpm (b) 25406.38 rpm (c) 5.236 hp, 0.29 hp]

$V = 220V \quad I_{a1} = 38A \quad N_1 = 600$ rpm $\quad R = 0.6\Omega \quad V_b = 3$

$E_{b1} = 220 - (38 \times 0.6) - 3 = 194.2V$

(a) $E_{b2} = 220 - (19 \times 0.6) - 3 = 205.6V$

$$N_2 = 600 \times \frac{205.6}{194.2} \times \frac{38}{19} = 1270.44$$

(b) $E_{b3} = 220 - (1 \times 0.6) - 3 = 216.4\,V$

$$N_3 = 600 \times \frac{216.4}{194.2} \times \frac{38}{1} = 25406.3 \qquad (746W = 1hp)$$

(c) $P_2 = E_{b2} \times I_{a2} = 205.6 \times 19 = 3906.4W = 5.236\,hp$

$P_3 = E_{b3} \times I_{a3} = 216.4 \times 1 = 216.4W = 0.29\,hp$

3. A 440 V series motor has a total series resistance (including field) of 0.8 Ω.

It takes 110 A to give the rated output at 800 rpm.

Determine the resistance to be added to obtain rated torque: $\Rightarrow I_2 = I_1 = 110A$

(i) At starting (ii) At 600 rpm.

Assume that the flux is proportional to the current.

$E_{b1} = 440 - (110 \times 0.8) = 352V$

(i) at starting : $E_b = 0 \Rightarrow V = 110(0.8 + R_{ext}) = 440$

$\Rightarrow R = 3.2\,\Omega$

(ii) $\frac{E_{b2}}{352} = \left(\frac{I_2}{I_1}\right)\left(\frac{N_1}{N_2}\right) = \frac{3}{4} \Rightarrow E_{b2} = 264\,V$

$440 = 264 + 110(0.8 + R_{ext}) \Rightarrow R_{ext} = 0.8\,\Omega$

4. A separately excited DC generator has no-load voltage of 120 V at a field current of 2 A, (E_{g1}, I_{f1})

when driven at 1500 rpm. Assuming that it is operating on the straight line portion of its (N_1)

saturation curve, calculate:

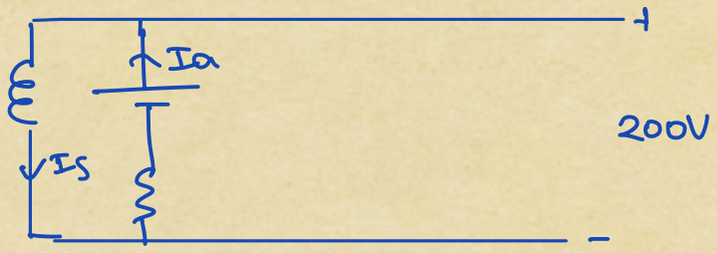
(a) The generated voltage when the field current is increased to 2.5 A.

(b) The generated voltage when the speed is reduced to 1400 rpm and the field current is

increased to 2.84 A.

Generated voltage $E_g = K I_f N$ (I_f = field current)

(a) $E_{g2} = 120 \times 1.25 = 150V$ (b) $\frac{14}{15} \times 1.42 \times 120 = 159.04V$



$198 = K(52.36) - 0.1 I_a$

$200 = I_s \cdot 100 \Rightarrow I_s = 2A$

$60 \times 10^3 = 200 \times (2 + I_a)$

$I_a = 302A$

$\Rightarrow K = 4.43$

5. A DC 60 kW shunt wound generator running at 500 rpm is supplying full load to a bus bar at 200 V. If the prime-mover input to the generator is removed and the machine now operates taking 5 kW from the bus bar, at what speed will it operate? The armature and field resistances are 0.1 Ω and 100 Ω respectively. Brush contact drop may be taken as 2 V. Neglect armature reaction.

[Ans: 421.4 rpm]

$198 = 4.43\omega - 0.1 \times 23$

6. A DC shunt generator with an armature resistance of 0.75 Ω and a field resistance of 125 Ω supplies a load of 10 kW at a terminal voltage of 250 V. Determine the efficiency of the generator assuming the iron losses, friction losses and windage losses amount to 600 W.

[Ans: 80.5%]

$I_s = 2A$

$I_a = 42A$

$P_{out} = 10K$

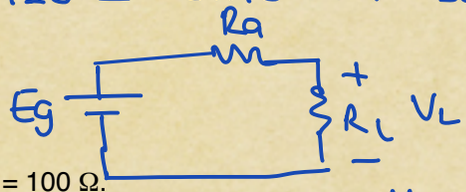
$P_{in} = 10K + 600 + I_a^2 R_a + I_s^2 R_s$

7. A separately excited DC generator is rated 6 kW, 120 V, 1200 rpm, and has the following open circuit characteristics at rated speed.

If (A) 0 0.1 0.2 0.3 0.4 0.5 0.6 0.8 1.0 1.2

Eg (V) 5 20 40 60 79 93 102 114 120 125

$120 \cdot I = 6 \times 10^3 \Rightarrow I = 50A$ (rated)



The machine parameters are Ra = 0.2 Ω and Rf = 100 Ω.

The machine is driven at 1200 rpm and the field current is adjusted at 0.8 A.

A load resistance, RL = 2 Ω is connected to the armature terminals.

Neglect armature reaction effect.

(a) Determine the quantity ke for the machine.

(b) Determine the generated voltage and the load current.

(c) Calculate the electrical power supplied to the load.

[Ans: (a) 0.9071 V/rad/s (b) 114 V (c) 5370.25 W]

$\frac{114}{2.2} = I_a \rightarrow I_a = 51.82$

$\Rightarrow V_t = 103.64$

$K_e = \frac{114}{\frac{20}{60} \times 1200} = 0.907$

$P_{load} = V_t I_a = 5370.25$

5. A DC 60 kW shunt wound generator running at 500 rpm is supplying full load to a bus ba

at 200 V. If the prime-mover input to the generator is removed and the machine now

operates taking 5 kW from the bus bar, at what speed will it operate?

The armature and field resistances are 0.1 Ω and 100 Ω respectively.

Brush contact drop may be taken as 2 V. Neglect armature reaction.

load current: $\frac{60\times10^3}{200} = 300A$

shunt current: $\frac{200}{100} = 2A \Rightarrow I_a = 302A$

$E_g = V + I_a R_a + V_{brush} = 200 + 30.2 + 2 = 232.2V$

$P_{in} = 5000W \Rightarrow I_{line} = 5000/200 = 25A$

$I_a = 25 - 2 = 23A \Rightarrow E_b = 200 - (23\times0.1) - 2 = 195.7V$

$\frac{N_2}{N_1} = \frac{E_b}{E_g} \Rightarrow N_2 = 500\times\frac{195.7}{232.2} = 421.4 rpm$

6. A DC shunt generator with an armature resistance of 0.75 Ω and a field resistance of

125 Ω supplies a load of 10 kW at a terminal voltage of 250 V.

Determine the efficiency of the generator assuming the iron losses, friction losses

and windage losses amount to 600 W.

$I_a = 42A$ $P_{a\text{-}loss} = 42^2\times0.75 = 1323W$

$P_{sh.loss} = 250\times2 = 500W$

total loss $= 1323 + 500 + 600 = 2423W$

$\eta = \frac{10000}{12423} = 80.5\%$

7. A separately excited DC generator is rated 6 kW, 120 V, 1200 rpm, and has the following

open circuit characteristics at rated speed.

If (A) 0 0.1 0.2 0.3 0.4 0.5 0.6 0.8 1.0 1.2

Eg (V) 5 20 40 60 79 93 102 114 120 125

The machine parameters are Ra = 0.2 Ω and Rf = 100 Ω.

$\omega = 125.66 rad/s$

$E_g = K_e \phi \omega \simeq K_e \omega$

a) $K_e = \frac{E_g}{\omega} = \frac{114}{125.6} = 0.907 \frac{Vs}{rad}$

The machine is driven at 1200 rpm and the field current is adjusted at 0.8 A.

A load resistance, RL = 2 Ω is connected to the armature terminals.

Neglect armature reaction effect.

(a) Determine the quantity ke for the machine.

(b) Determine the generated voltage and the load current.

(c) Calculate the electrical power supplied to the load.

b) $E_g = I_a(R_a + R_L)$

$114 = I_a(2.2) \Rightarrow I_a = 51.82\,A$

c) $P = 51.82^2 \times 2 = 5370.6\,W$

EE2005: Electrical Machines
Tutorial 6

1. The full load torque angle of a synchronous motor at rated voltage and frequency is 35° (electrical). Neglect the effects of armature resistance and leakage reactance. If the field current is held constant, how would the full-load torque angle be affected by the following changes in operating condition?

   (a) Frequency reduced by 10 %, load torque and applied voltage constant.

   (b) Frequency reduced by 10 %, load power and applied voltage constant.

   (c) Both frequency and applied voltage reduced by 10%, load torque constant.

   (d) Both frequency and applied voltage reduced by 10%, load power constant.

   [**Ans:** (a) $\delta = 31.07°$ (b) unchanged (c) unchanged (d) $\delta = 39.6°$]

2. A three phase star connected alternator is synchronised with an infinite bus of 11 kV. The prime mover input is then increased such that the output power reaches 15 MW. Now, the field excitation is increased to 130%, when the machine starts operating at 0.8 pf lag. Neglecting the armature resistance, determine:

   (a) synchronous reactance of the machine.

   (b) power factor, load angle and armature current before the field excitation was increased.

   [**Ans:** (a) 2.7406 Ω (b) 0.985 (lead), 19.86°, 800.4 A]

3. A three phase synchronous machine with $X_s = 0.9$ pu on its own rating, is connected via a transmission line to an infinite bus whose voltage is 1 pu. The line reactance is 0.15 pu on the machine base. The internal emf E_f of the machine is kept at 1.35 pu. Calculate

   (a) pull out power of the machine when operating as generator.

   (b) generator terminal voltage and current at pull out.

   (c) reactive powers at the terminals of the generator and at the infinite bus at the moment of pull out.

   [**Ans:** (a) 1.286 pu (b) 0.8786 pu, 1.6 pu (c) -0.568 pu, -0.952 pu]

4. A Y-connected synchronous generator connected to an infinite bus at 12 kV supplies 4000 A at unity power factor. The synchronous reactance is 2 Ω per phase and the stator resistance is negligible. If the prime mover set point is unchanged and the field current is increased by 15 %, determine the new stator current and power factor.

   [**Ans:** 4154 A, 0.9628 (lag)]

1. The full load torque angle of a synchronous motor at rated voltage and frequency is 35° (electrical). Neglect the effects of armature resistance and leakage reactance. If the field current is held constant, how would the full-load torque angle be affected by the following changes in operating condition?

   (a) Frequency reduced by 10 %, load torque and applied voltage constant.

   (b) Frequency reduced by 10 %, load power and applied voltage constant.

   (c) Both frequency and applied voltage reduced by 10%, load torque constant.

   (d) Both frequency and applied voltage reduced by 10%, load power constant.

$T = \frac{EV}{X_S \omega_S} \sin\delta$ $\quad E_\phi \propto f_S$ (supply frequency)

$\omega_S = \frac{2\pi N_S}{60} = \frac{2\pi}{60} \times \frac{120 f_S}{P} \Rightarrow \omega_S \propto f_S$

$X_S = \omega L = 2\pi f_S L \Rightarrow X_S \propto f_S$

$P \propto \frac{EV}{f_S} \sin\delta$

$T \propto \frac{EV}{f_S^2} \sin\delta$

a) $f_2 = 0.9 f_1 \quad E_2 = 0.9 E_1 \quad \delta_2 = \sin^{-1}(0.9 \times \sin 35) = 31.08^\circ$

b) 35°

c) $f_2 = 0.9 f_1 \quad V_2 = 0.9 V_1 \quad E_2 = 0.9 E_1 \quad 35^\circ$

d) $\sin^{-1}\left(\frac{1}{0.9} \sin 35\right) = 39.59^\circ$

2. A three phase star connected alternator is synchronised with an infinite bus of 11 kV. The prime mover input is then increased such that the output power reaches 15 MW. Now, the field excitation is increased to 130%, when the machine starts operating at 0.8 pf lag. Neglecting the armature resistance, determine:

   (a) synchronous reactance of the machine.

   (b) power factor, load angle and armature current before the field excitation was increased.

$E_\phi = 1.3 V_\phi$

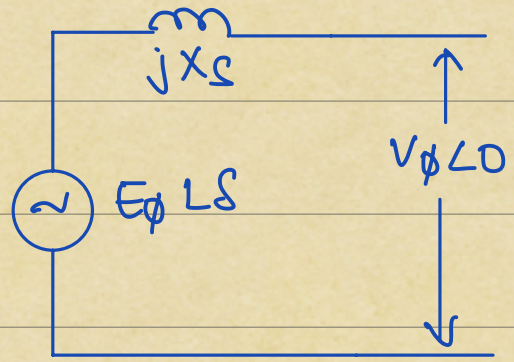
$V_\phi = 11 \times 10^3 / \sqrt{3}$

$P = \frac{|E||V|}{X_S} \sin\delta = \frac{3|E_\phi V_\phi|}{X_S} \sin\delta$

$E_\phi \angle\delta = \frac{E}{\sqrt{3}} \angle\delta$

$V_\phi \angle\delta = V/\sqrt{3} \angle\delta$

a)



$\sqrt{3} V_L I_L \cos\phi = 15MW$

$\Rightarrow I_L = \frac{15 \times 10^6}{\sqrt{3} \times 11 \times 10^3 \times 0.8} = 984.12A$

$E\phi^2 = (V\phi \cos\phi + I_a \cancel{R_a}^{\to 0})^2 + (V\phi \sin\phi + I_a X_s)^2$

$\Rightarrow E\phi^2 = V\phi^2 + (I_a X_s)^2 + 2V\phi \sin\phi \, (I_a X_s)$

$\Rightarrow I_a X_s = 2697.18V \Rightarrow X_s = 2.74$

b) $P_o = \frac{EV}{X_s} \sin\delta \Rightarrow 15 \times 10^6 = \frac{11 \times 10^3 \times 11 \times 10^3}{2.74} \sin\delta \Rightarrow \delta = 19.862^\circ$

$I_a = \frac{E\phi - V\phi}{jX_s} = \frac{|V\phi|}{X_s} \frac{\angle 19.862^\circ - 0^2}{\angle 90^\circ} = 799.27 \angle 9.931$

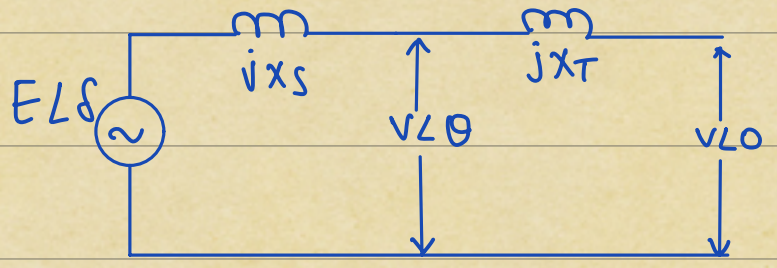
PF = cos (9.931) = 0.985 lead

3. A three phase synchronous machine with $X_s = 0.9$ pu on its own rating, is connected via a transmission line to an infinite bus whose voltage is 1 pu. The line reactance is 0.15 pu on the machine base. The internal emf E_f of the machine is kept at 1.35 pu. Calculate

   (a) pull out power of the machine when operating as generator.

   (b) generator terminal voltage and current at pull out.

   (c) reactive powers at the terminals of the generator and at the infinite bus at the moment of pull out.



a) $P_o = \frac{EV}{X_s + X_T} = \frac{1.35 \times 1}{0.9 + 0.15} = 1.29 \, pu.$

b) $I = \frac{E\angle 90 - V\angle 0}{1.05j} = \frac{1.35\angle 90 - 1}{1.05\angle 90} = 1.6\angle 36.53$ p.u.

$V_t = V + I(0.15) = 1\angle 0 + 1.6\angle 36.53 \times (0.15)$

$= 0.88\angle 12.68$ p.u

c) $S = V_t I^* = 0.88\angle 12.68 \times 1.6\angle -36.53$

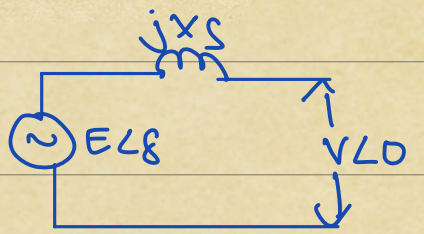
$= 1.28 - j\underbrace{0.57}_{Q_{gen}}$

$S = VI^* = 1\angle 0 \times 1.6\angle -36.53 = 1.28 - j\underbrace{0.95}_{Q_{bus}}$

4. A Y-connected synchronous generator connected to an infinite bus at 12 kV supplies 4000 A at unity power factor. The synchronous reactance is 2 Ω per phase and the stator resistance is negligible. If the prime mover set point is unchanged and the field current is increased by 15 %, determine the new stator current and power factor.

$X_S = 2\Omega$

$V_\phi = \frac{12}{\sqrt{3}} \times 10^3 \angle 0 \qquad I = 4000\angle 0$



$E_\phi\angle\delta = \frac{12}{\sqrt{3}} \times 10^3 + 4000\angle 0\,(2j)$

$= 10583\angle 49.1^\circ$ V $\qquad I_f \propto |E|$

$I \Rightarrow 1.15I \Rightarrow E_{2\phi} = 1.15|E_\phi|\angle\delta_2 = 12170.46\angle\delta_2$

$\left|\frac{E_\phi V}{X_S}\right| \sin\delta_1 = \left|\frac{E_{2\phi} V}{X_S}\right| \sin\delta_2$

$\Rightarrow \delta_2 = 41.096^\circ \Rightarrow E_{2\phi} = 12170.46\angle 41.1^\circ$

$I_2 = \frac{E_{2\phi} - V_\phi}{jX_S} = 4154.31\angle -15.67^\circ$

PF = 0.96 lagging

5. A 4-pole, 60 Hz, 24 kV, 650 MVA synchronous generator with a synchronous reactance of 1.82 per unit is connected to a grid which can be represented by a 24 kV infinite bus in series with a reactive impedance of j 0.21 Ω. The generator is equipped with a voltage regulator that adjusts the field excitation such that the generator terminal voltage remains at 24 kV independent of the generator loading. The output power of the generator is adjusted to 375 MW.

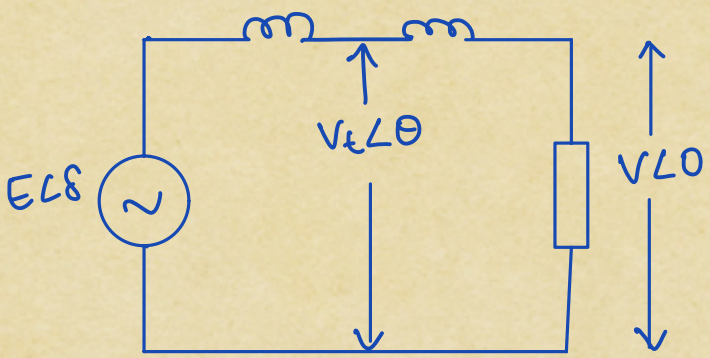
   (a) Draw a phasor diagram for this operating condition.

   (b) Find the magnitude and phase angle (with respect to the generator terminal voltage) of the generator terminal current.

   (c) Determine the generator terminal power factor.

   (d) Find the magnitude (in p.u and kV) of the generator excitation voltage E_f .

   [**Ans:** (b) 0.578 pu, 3.93° (c) 0.9976 (lag) (d) 1.5 pu, 36 kV]



$650 = \sqrt{3} V_L I_L \Rightarrow I_L = \frac{650M}{\sqrt{3} \cdot 24 \times 10^3} = 15636A$

$Z_{base} = \frac{V^2}{S} = \frac{(24 \times 10^3)^2}{650 \times 10^6} = 0.221\ pu$

EE2005: Electrical Machines
Tutorial 8

1. Consider a stator coil of N turns, with the sides of the coil separated 180° apart. The length of the coil side is l. The coil links a flux density waveform produced by the rotor which varies as $B(\phi) = B_m \cos(\phi)$, where ϕ is the angle around the circumference of the stator. With respect to this waveform, the coil sides may be taken to be at angles $\phi = 0°$ and $\phi = 180°$ and the rotor field axis positioned at $\phi = 0^o$. With respect to this waveform, the coil sides may be taken to be at angles $\phi = 0^o$ and $\phi = 180^o$.

   (a) If this flux density waveform now rotates in the air gap due to the rotation of the rotor, find the expression for B when the rotor is at some angle $\theta = \omega t$. $B_m \cos(\phi - \omega t)$

   (b) Derive an expression for the flux linkage of the stator coil at any angle θ of the rotor. (Hint: Find the flux passing through an elemental area at an angle ϕ, and integrate to find the total flux passing through the area spanned by the coil).

2. Now consider another identical coil placed in the adjacent slot, which is situated at an angle of δ away from the first, i.e the coil sides may be assumed to lie at angles $\phi = \delta$ and $\phi = \pi + \delta$. Derive the expression for the voltage induced in that coil. If the two coils are now connected in series, what would be the net voltage?

3. A 208 V, 10 hp, 4 pole, 60 Hz, Y-connected induction motor has a full-load slip of 5 %.

   $N_S = 120f/P = 1800$ RMP $0.05 = \frac{N_S - N_r}{N_S} \Rightarrow N_r = 1710$ RPM

   (a) What is the synchronous speed of this motor?

   (b) What is the rotor speed of this motor at the rated load?

   (c) What is the rotor frequency of this motor at the rated load? $f_r = sf = 3$ Hz

   (d) What is the shaft torque of this motor at the rated load? $\frac{P_{out}}{\omega_r} = \frac{10 \times 746}{2\pi \times 1710/60} = 41.64$ NM

4. A 460 V, 25 hp, 60 Hz, 4-pole, Y-connected induction motor has the following impedances in ohms per phase referred to the stator circuit:
   $R_1 = 0.641\,\Omega$, $R_2' = 0.332\,\Omega$, $X_1 = 1.106\,\Omega$, $X_2' = 0.464\,\Omega$, $X_m = 26.3\,\Omega$
   The total rotational losses are 1100 W and are assumed to be constant. The core loss is lumped in with the rotational losses. For a rotor slip of 2.2% at the rated voltage and rated frequency, find the motor's

   (a) Speed

   (b) Stator current

   (c) Power factor

   (d) Resistance equivalent to mechanical load

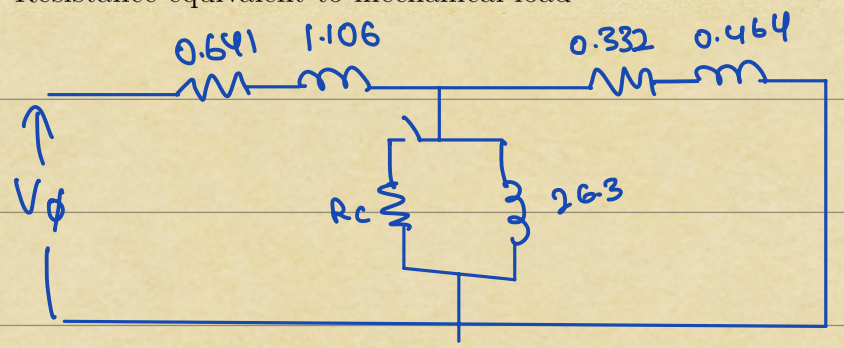
5. The following results were obtained on a three phase, 75 kW, 3.3 kV, 6-pole, 50 Hz, Y-connected squirrel cage induction motor.
   No load Test: $f = 50$ Hz, $V_0 = 3.3$ kV, $I_0 = 5$ A, $P_0 = 2500$ W

4. A 460 V, 25 hp, 60 Hz, 4-pole, Y-connected induction motor has the following impedances in ohms per phase referred to the stator circuit:
$R_1 = 0.641\,\Omega$, $R_2' = 0.332\,\Omega$, $X_1 = 1.106\,\Omega$, $X_2' = 0.464\,\Omega$, $X_m = 26.3\,\Omega$
The total rotational losses are 1100 W and are assumed to be constant. The core loss is lumped in with the rotational losses. For a rotor slip of 2.2% at the rated voltage and rated frequency, find the motor's

(a) Speed
(b) Stator current
(c) Power factor
(d) Resistance equivalent to mechanical load

$N_S = \frac{120 \times 60}{4} = 1800$ rpm

a) $N_r = (1 - 0.022) N_s$ 1760.4 rpm



b) $Z_{eq} = 14.06 \angle 33.68^\circ\ \Omega$

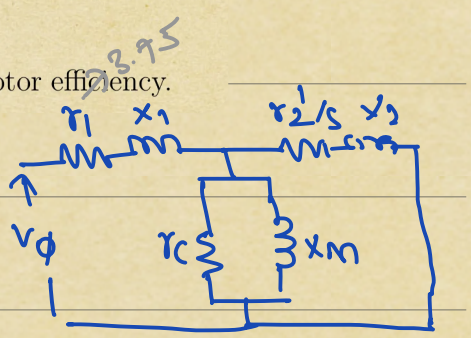
$I = \frac{460/\sqrt{3}}{Z_{eq}} = 18.89 \angle -33.68^\circ$

c) $PF = \cos(-33.68) = 0.83$ lag

d) $\frac{R_2'(1-s)}{s} = 14.759\ \Omega$

5. The following results were obtained on a three phase, 75 kW, 3.3 kV, 6-pole, 50 Hz, Y-connected squirrel cage induction motor.
No load Test: $f = 50$ Hz, $V_0 = 3.3$ kV, $I_0 = 5$ A, $P_0 = 2500$ W
Blocked rotor Test: $f = 15$ Hz, $V_{BR} = 400$ V, $I_{BR} = 27$ A, $P_{BR} = 15$ kW
The stator resistance per phase is 3.75 Ω.

(a) Determine the parameters of the circuit model.
(b) For a slip of 0.04, calculate the stator current, its power factor and motor efficiency.



a) $V_\phi = \frac{3.3\text{kV}}{\sqrt{3}}$ $I_0 = 5$A $P_0 = 2500$W

$2500\text{W} = P_{iron} + P_{mech} + P_{stator\ loss}$

$= P_{iron} + 3I_0^2 R_1$

$2500 = P_{Fe} + 3(5)^2(3.75) \Rightarrow P_{Fe} = 2218\text{W} = \sqrt{3} V_0 I_c$

Blocked rotor Test: $f = 15\,\text{Hz}$, $V_{BR} = 400\,\text{V}$, $I_{BR} = 27\,\text{A}$, $P_{BR} = 15\,\text{kW}$
The stator resistance per phase is $3.75\,\Omega$.

(a) Determine the parameters of the circuit model.

(b) For a slip of 0.04, calculate the stator current, its power factor and motor efficiency.

EE2005: Electrical Machines
Tutorial 10

1. A 2 - pole, 50 Hz induction motor supplies 15 kW to a load at a speed of 2950 rpm.

   (a) What is the motors slip?

   (b) What is the developed torque in the motor in Nm under these conditions?

   (c) What will the operating speed of the motor be if its torque is doubled?

   (d) How much power will be supplied by the motor when the torque is doubled?

   [**Ans:** (a) 1.67% (b) (b) 48.6 Nm (c) 2900 rpm (d) 29.5 kW]

2. A 460 V, 25 hp, 60 Hz, 4-pole, Y-connected induction motor has the following impedances in ohms per phase referred to the stator circuit:
   $R_1 = 0.641\,\Omega$, $R_2' = 0.332\,\Omega$, $X_1 = 1.106\,\Omega$, $X_2' = 0.464\,\Omega$, $X_m = 26.3\,\Omega$
   The total rotational losses are 1100 W and are assumed to be constant. The core loss is lumped in with the rotational losses. Consider the induction motor in Q.2 ,

   (a) What is the maximum torque of this motor? At what speed and slip does it occur?

   (b) What is the starting torque of this motor?

   (C) When the rotor resistance is doubled, what is the speed at which the maximum torque now occurs? What is the new starting torque of the motor?

   [**Ans:** (a) 240.185 Nm, 1447.56 rpm, slip = 0.1958 (b) 109.24 Nm (c) 1095.12 rpm, 178.83 Nm]

3. A 460 V, 1740 rpm, 60 Hz, 4-pole, wound rotor induction motor has the following parameters per phase:
   $R_1 = 0.25\,\Omega$, $R_2' = 0.2\,\Omega$, $X_1 = 0.5\,\Omega$, $X_2' = 0.5\,\Omega$, $X_m = 30\,\Omega$
   The rotational losses are 1700 W. With the rotor terminals short-circuited, find

   (a) (i) Starting current when started direct on full voltage.
   (ii) Starting torque.

   ((b) (i) Full-load slip.
   (ii) Full-load current.
   (iii) Ratio of starting current to full-load current.
   (iv) Full-load power factor.

   Assume approximate equivalent circuit.

   [**Ans:** (a) (i)250.29 $\angle$ -66.66° A (ii) 186.7 Nm (b) (i) 0.03333 (ii)43.83 $\angle$ -20.51° A (iii) 4.259 (iv) 0.93 lag]

4. A three-phase, 460 V, 60 Hz, six-pole wound-rotor induction motor drives a constant load of 100 Nm at a speed of 1140 rpm when the rotor terminals are short-circuited. It is required to reduce the speed of the motor to 1000 rpm by inserting resistances in the rotor circuit. Determine the value of the resistance if the rotor winding resistance per phase is 0.2 Ω. Neglect rotational losses. The stator-to-rotor turns ratio is unity.

   [**Ans:** 0.468 Ω per phase]

5. A 6-pole, 50 Hz, 3-phase induction motor running on full load develops a useful torque of 162 Nm, and it is observed that the rotor emf makes 90 complete cycles per min. Calculate the power output. If the mechanical torque lost in friction be 13.5 Nm, find the copper loss in rotor windings, the input to the motor and efficiency. The stator losses are 750 W.

   [**Ans:** Rotor copper loss = 551.35 W, Efficiency = 86.02 %]

6. A 400 V, 50 Hz, 30 hp, 3-ϕ induction motor is drawing 50 A current at 0.8 power factor lagging. The stator and rotor copper losses are 1.5 kW and 900 W respectively. The friction and windage losses are 1050 W and the core losses are 1200 W. Calculate the air-gap power of the motor.

   [**Ans:** 25 012.8 W]

7. The blocked rotor test on a 37 kW, 440 V, 50 Hz, 6-pole, Y-connected, 3-ϕ induction motor gave the following results: $V_{line} = 230$ V , $I_{line} = 140$ A, $\cos\phi = 0.35$. The motor drives a load having a constant torque of 300 Nm. Estimate the maximum percentage reduction in supply voltage possible before the motor stalls. Assume that the copper losses are equally divided between the stator and the rotor. Neglect magnetising current.

   [**Ans:** 41.1 %]

8. An 8-pole, 50 Hz, 3-ϕ induction motor has an equivalent rotor resistance of 0.07 Ω/phase. If its stalling speed is 630 rpm, how much resistance is to be included per phase to obtain the maximum torque at starting? Ignore magnetising current.

   [**Ans:** 0.3675 Ω]

9. An induction motor has an effiency of 0.9 when the load is 50 hp. At this load, the stator and rotor copper loss each equal the iron loss. The mechanical losses are one-third of the no-load loss. Find the slip.

   [**Ans:** 0.0303]

10. A 400 V, 50 Hz, 4-pole, 1400 rpm, Y-connected squirrel cage induction motor has the following parameters referred to the stator: $R_2' = 1.0\,\Omega$, $X_1 = X_2' = 1.5\,\Omega$. Neglect stator resistance and core and rotational losses of the motor. The motor is controlled from a 3-phase voltage source inverter with constant V/f control. Calculate the stator voltage and frequency to obtain the maximum torque at starting. Use the approximate equivalent circuit.

    [**Ans:** 16.7 Hz, 133.33 V]

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EE 2005: Electrical Machines

Tutorial Sheet on 3 Phase transformers and p.u

1. A 3-phase transformer is used to step down the voltage from 3300 V supply mains. The turns ratio per phase of the transformer is 10. If the load is adjusted such that it draws 10 A of current from the supply mains in either of the two cases below, determine the secondary line voltage, line current and output kVA when the transformer windings are connected in (a) star-delta (b) delta-star.

   Y/Δ: 190.526V, 173.21A, 57.15 kVA. Δ/Y : 571.576 V, 57.73A, 57.15 kVA

2. A 100 MVA, 230/115 kV delta – delta connected 3 phase power transformer has a resistance of 0.02 p.u and a reactance of 0.055 pu. Find the voltage regulation of this transformer when it is supplying an inductive load of 80 MVA at 115 kV and 0.85 power factor.

   input voltage is 3.718% higher

3. A 150 kVA 3 phase transformer is rated 2400/460 V and has a series impedance of 0.038 + j 0.135 p.u on its own base. It supplies power to a 3-phase balanced load drawing 95 kVA at unity power factor and at a terminal line voltage of 438 V. Calculate the line voltage at the HV side of the transformer. Do the calculations on a 460 V, 100 kVA base. 2355.74 V

4. A delta-star connected bank of three identical 100 kVA 2400/120 V, 50 Hz transformers is supplied with power through a feeder whose impedance is 0.3+j0.8 Ω per phase. The voltage at the sending end of the feeder is held constant at 2400 V line to line. The results of a single-phase short circuit test on one of the transformers with its low-voltage terminals short-circuited are V_H=57.5 V, f=50 Hz, I_H=41.6 A, P=875 W.
   a. Determine the secondary line to line voltage when the bank delivers rated current to a balanced three phase unity pf load.
   b. Compute the currents in the transformer primary and secondary windings and in the feeder wires if a solid three phase short circuit occurs at the secondary line terminals.

5. Three Single phase transformers each rated 25 kVA, 3300/300 V are delta – delta connected to form a 3 phase bank. It is used to supply a load of 50 kVA at the secondary side. (a) Calculate the current in the primary winding of each transformer. During operation, one of the transformers develops a fault and is removed from service, with the remaining transformers used to supply the load in an open delta configuration. (a) Calculate the kVA supplied by each transformer and the total kVA rating of the set. (b) If the load has a power factor of 0.95 lagging, calculate the power factor of operation of each transformer. What would be the power factor when the load is purely resistive?

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